



Article 000006815

The licensing features available on Ruckus Switches

Summary

Article has got information about the different types of licensing and their features available on Ruckus Switches

Question

What are the licensing features available on Ruckus Switches?

Customer Environment

Wired Switch setup ICX 8200 ICX 7850 ICX 7750 ICX 7650 ICX 7250 ICX 7450 ICX 7150 ICX 6450 ICX 6610 ICX 6430

Workaround

Please contact [Ruckus Customer Support](#) for further assistance

Resolution

ICX 6610 License Features

The ICX 6610 device supports the use of licensed FastIron feature or feature sets.
The following feature licenses are available for the ICX 6610 device:

Premium Layer 3 license
Ports on Demand license
MACSec license

License Part number/SKU	Function
ICX-MACSEC-LIC	This license enables MACSec functionality for any 10 Gbps port on an ICX 6610 device
ICX6610-10G-LIC-POD	License to upgrade 4 ports of 1 GbE SFPP uplink to 10 GbE. The port capacities for this license are 4 ports or 8 ports.
ICX6610-PREM-LIC	ICX6610 Premium software license (includes BGP, VRF)

Premium Layer 3 license

This license enables you to use the following features or feature sets:

BGP4
BGP4+
GRE
IPv6 static routing
IPv6 unicast routing
IPv6 over IPv4 tunnels
OSPF v2
OSPF v3
PIM-DM
PIM-SM
PIM-SSM
PIM passive
PBR
RIPng
VRRP

VRRP v3 for IPv6
VRRP-E
VRRP-E for IPv6
VRF-Lite

ICX6430: Does not have any additional license features

ICX 6450 and ICX 6450C License Features

The ICX 6450 and ICX 6450C devices support the use of licensed FastIron feature or feature sets.

License Part number/SKU	Function
ICX6450-PREM-LIC	Ruckus ICX 6450/6450-C premium license (Layer 3 features)
ICX6450-2X10G-LIC-POD	Ruckus ICX 6450 2x10 GbE capacity-based license; upgrade 1 GbE uplink/stacking ports to 1 GbE/10 GbE

Premium Layer 3 license

This license is available for the ICX 6450 and ICX 6450C devices. It enables you to use the following features or feature sets:

IPv4 OSPF v2 (ICX 6450 only)
IPv4 OSPF v3 (ICX 6450 only)
RIP v1, v2 (ICX 6450 only)
RIPng (ICX 6450 only)
VRRP (ICX 6450 and ICX 6450C)
VRRP-E (ICX 6450 and ICX 6450C)
GRE (ICX 6450 and ICX 6450C)

FCX License Features

The FCX device supports the use of licensed FastIron feature or feature sets.

Advance Layer 3 license

This license enables you to use the following features or feature sets:

- BGP
- BGP4+
- GRE
- IPv6 over IPv4 tunnels

FSX 800 and FSX 1600 License Features

The FSX 800 and FSX 1600 devices support the use of licensed FastIron feature or feature sets.

IPv4 and IPv6 PREM Layer 3 license (XL management modules)

This license enables you to use the following features or feature sets:

- BGP
- BGP4+
- IPv6 Layer 3 forwarding
- IPv6 over IPv4 tunnels in hardware
- IPv6 redistribution
- IPv6 static routes
- RIPng
- IPv6 VRRPv3
- IPv6 over GRE
- PIM-DM (third generation only)
- IPv6 PIM-SM (third generation only)
- IPv6 PIM passive (third generation only)

Pv6 PIM SSM (third generation only)
IPv6 Policy-based routing (third generation only)
IPv6 over IPv4 tunnels
6,000 active host routes
OSPF v2
OSPF v3
Anycast RP
DVMRP V2
IGMP V1, V2, and V3
ICMP redirect messages
IGMP V3 fast leave (for routing)
MSDP
PIM-DM
PIM-SM
PIM passive
PIM SSM
Policy-based routing
VRRP-E
GRE
Multicast over GRE
IPv4 over GRE
Multi-VRF

NOTE: The licenses for FCX, FSX 800 and FSX 1600 are not on sale anymore as they are End of Life models.

ICX 7250 License Features

The ICX 7250 device supports the use of licensed FastIron feature or feature sets.
The following feature licenses are available for the ICX 7250 device:

Premium Layer 3 (non-node locked license)

Ports on Demand (PoD) license

License Part number/SKU	Function
ICX7250-PREM-LIC	License to upgrade any Ruckus ICX 7250 (except ICX7250-24G) to Layer 3 Premium Features (OSPF, VRRP, PIM, PBR, VRF), non-node lock license
ICX7250-2X10G-LIC-POD	License to upgrade any Ruckus ICX 7250 (except ICX7250-24G) from 8×1 GbE SFP to 6×1 GbE SFP and 2×10 GbE SFP+ uplink/stacking ports, node-lock license
ICX7250-8X10G-LIC-POD	License to upgrade any Ruckus ICX 7250 (except ICX7250-24G) from 6×1 GbE SFP and 2×10 GbE SFP+ to 8×10 GbE SFP+ uplink/stacking ports, node-lock license. Only switches that already have 2×10 GbE ports can be upgraded to 8×10 GbE ports.

Premium Layer 3 (non-node locked license)

This license enables you to use the following features or feature sets:

- OSPF v2
- RIP v1/v2
- VRRP
- VRRP-E
- VRF (IPv4 and IPv6)
- IPv6 over IPv4 Tunnels

ICX 7450 License Features

The ICX 7450 device supports the use of licensed FastIron features or feature sets. The following feature licenses are available for the ICX 7450 device:

Premium Layer 3 (non-node locked license) MACSec license

License Part number/SKU	Function
ICX7450-PREM-LIC	Ruckus ICX 7450 Layer 3 Premium Software License
ICX-MACSEC-LIC	License to enable MACsec encryption.

This license enables you to use the following features or feature sets:

- OSPF v2
- OSPF v3 (IPv6)
- RIP v1/v2
- RIPng (IPv6)
- VRRP
- VRRPv3 (IPv6)
- VRRP-E
- GRE
- PBR
- PIM-SM, PIM-SSM, PIM-DM
- PIM Passive
- BGP, BGP4+ (IPv6)
- VRF (IPv4 and IPv6)
- GRE
- IPv6 over IPv4 Tunnels

ICX 7650 License Features

The ICX 7650 device supports the use of licensed FastIron feature or feature sets.
The following feature licenses are available for the ICX 7650 device:

Premium Layer 3 (non-node locked license) MACSec license

License Part number/SKU	Function
ICX7650-PREM-LIC	ICX 7650 advanced L3 License adds support for OSPF, BGP VRRP, PIM, PBR, VRF
ICX-MACSEC-LIC	ICX MACsec license adds support for 128/256 bit MACsec encryption to ICX 7650

Premium Layer 3 IP routing (with software license)

- IPv4 and IPv6 dynamic routes
- OSPF v2, OSPF v3 (IPv6)
- PIM-SM, PIM-SSM, PIM-DM, PIM passive (IPv4/IPv6 multicast routing functionality)
- PBR
- Virtual Route Redundancy Protocol VRRP v3 (IPv6)
- VRRP-E (IPv4, IPv6)
- BGP4, BGP4+ (IPv6)
- GRE
- IPv6 over IPv4 tunnels
- VRF-lite (IPv4 and IPv6)
- MSDP

ICX 7750 License Features

License Part number/SKU	Function
ICX7750-L3-COE	Certificate of Entitlement to use routing and advanced functionality. Without the Certificate of Entitlement, customers may use base Layer 3 features: VRRP, RIP, and static routes. Other Layer 3 features are considered advanced and require the ICX7750-L3-COE. The Certificate of Entitlement is serialized paper that is not tied to a particular switch; no activation is required.

Premium Layer 3 IP routing feature set (with Certificate of Entitlement)

- IPv4 and IPv6 dynamic routes
- OSPF v2, OSPF v3 (IPv6) • PIM-SM, PIM-SSM, PIM-DM, PIM passive (IPv4/IPv6 multicast routing functionality)
- PBR
- VRRP-E (IPv4, IPv6)
- VRRPv3 (IPv6)
- BGP4, BGP4+ (IPv6)
- GRE
- IPv6 over IPv4 tunnels
- VRF (IPv4 and IPv6)

ICX 7850 License Features

The ICX 7850 device supports the use of licensed FastIron feature or feature sets.
The following feature licenses are available for the ICX 7850 device:

Premium Layer 3 (non-node locked license)
MACSec license

License Part Number / SKU :
ICX7850-PREM-LIC : ICX 7850 Layer 3 Premium software license

ICX-MACSEC-LIC : ICX MACSec License

Premium Layer 3 IP routing (with software license)

- IPv4 and IPv6 dynamic routes
- OSPF v2, OSPF v3 (IPv6)
- PIM-SM, PIM-SSM, PIM-DM, PIM passive (IPv4/IPv6 multicast routing functionality)
- PBR
- Virtual Route Redundancy Protocol VRRP v3 (IPv6)
- VRRP-E (IPv4, IPv6)
- BGP4, BGP4+ (IPv6)
- GRE
- IPv6 over IPv4 tunnels
- VRF-lite (IPv4 and IPv6)
- MSDP

ICX 8200 License Features

Premium Layer 3 IP routing feature set (with Certificate of Entitlement)

- IPv4 and IPv6 dynamic routes
- OSPF v2, OSPF v3 (IPv6) • PIM-SM, PIM-SSM, PIM-DM, PIM passive (IPv4/IPv6 multicast routing functionality)
- PBR
- VRRP-E (IPv4, IPv6)
- VRRPv3 (IPv6)
- BGP4, BGP4+ (IPv6)
- GRE
- IPv6 over IPv4 tunnels
- VRF (IPv4 and IPv6)

License Management

How-To Hub Licensing page

License Manager - LiMAN

License Purchases - Download Licenses

Temporary License

Asset and Case Management

Asset Management Tool

AP Certificate Replacement Tool

Hardware Summary

Case and RMA Management

Support Purchases - SPA Required

Activate Purchase - Purchase Code Required

Submit a Case

Cloud Subscription Purchases

Warranty & RMA

Register Warranty

Warranty Checker

"Support Service" Missing or Expiring

RMA Return Instructions

Knowledge Base

Advanced Search

Security Bulletins

Technical Documentation - Recent Releases

How-To Hub

RUCKUS Community Forums

Technical Support Bulletins

RUCKUS Education YouTube Channel

RUCKUS Networks University

Contact Us/Support Phone Numbers

Contact Us/Support Phone Numbers

Feedback

Ruckus Networks

350 West Java Dr.

Sunnyvale, CA 94089 USA

T: +1 (855) 478-2587

© 2026 Ruckus Wireless LLC. All Rights Reserved.